SOLUTION DESIGN DOCUMENT (SDD)

RPA solution • SAP Vendor Payment Process Automation

1. Design purpose

This document defines a practical technical design for an Automation Anywhere v11 bot that uses approved SAP recording-based interactions to generate a vendor-payment ledger report. It is a design baseline and must be reconciled with the actual SAP screens and enterprise standards before build or release.

2. Solution architecture

Component	Design
Scheduler / trigger	Manual or scheduled trigger, as approved by the process owner.
Automation Anywhere v11	Main task bot, reusable sub-tasks, exception handling and logging.
Credential store	Organization-approved credential vault; no credentials embedded in bot code.
SAP GUI	Target application accessed through approved SAP recording / GUI actions.
File system / shared location	Approved output folder with access controls and retention rules.
Run log	Automation log plus a business-friendly status record; avoid sensitive data.

3. High-level workflow

1. Start / load run parameters
2. Validate parameters and output folder
3. Check SAP availability and obtain credentials securely
4. Open SAP and navigate to approved report transaction
5. Apply date filters and selection criteria
6. Execute report and verify response

7. Export report and validate output

8. Write status / notify owner

9. End or route exception for review

4. Suggested bot modules

Module / task bot	Responsibility
Main_Controller	Coordinates execution, status and final cleanup.
Read_And_Validate_Inputs	Checks date range, mandatory parameters, output path and naming convention.
SAP_Login	Establishes SAP session using secure credentials; detects login failure.
Navigate_And_Run_Report	Uses validated SAP recording steps to open the report and execute it.
Validate_SAP_Response	Checks expected screen, report title/status, selected period and empty-result conditions.
Export_And_Save	Exports the report and verifies that the file exists and is non-empty.
Write_Run_Log	Records run ID, timestamps, status, parameters and sanitized error details.
Notify_Process_Owner	Sends success or exception notice through the approved channel, if configured.

5. Configuration and input data

Parameter	Type / rule	Validation
Run_ID	Generated unique text	Must be present in logs.
Start_Date	Date	Required; valid date; not after End_Date.
End_Date	Date	Required; valid date; consistent with business calendar rules.
Company_Code / selection	Text; optional or required per process	Must match approved SAP values.
Output_Folder	Path	Must exist or be creatable by the bot account and be writable.
File_Name	Text template	Must follow approved naming convention; avoid unsafe

		characters.
Run_Mode	Test / production	Production mode requires approved deployment and access.

6. SAP interaction design

- Prefer stable SAP GUI controls and verified screen identifiers where supported; avoid relying only on fixed screen coordinates.
- Use explicit waits and screen-state checks after login, navigation, report execution and export.
- Confirm transaction code, menu path, field IDs, required variants and export format with the SAP functional team.
- Detect SAP status-bar messages and modal dialogs; classify known safe messages separately from unexpected conditions.
- If a screen does not match the expected state, stop rather than entering data into an uncertain field.
- Use the least-privilege bot account and follow the organization's SAP and segregation-of-duties policies.

7. Data validation and output handling

- Validate that the requested date range is the same range shown in the generated report where visible.
- Confirm the report export completed, the file exists, and the file size is greater than zero.
- Where feasible, check required columns or report headings and record count without changing source data.
- Treat a zero-row report as a distinct business outcome requiring confirmation if zero results are not expected.
- Use a timestamp or run ID in the filename to prevent accidental overwrite unless a controlled overwrite rule is approved.
- Restrict output folder permissions to authorized finance users and retain files according to policy.

8. Error handling and retry policy

Error category	Handling strategy	Retry guidance
Input / configuration	Fail fast; log the missing or invalid parameter.	No automatic retry; correct input first.
SAP login / connectivity	Stop, log sanitized error and alert support.	Retry only under an approved bounded policy.
Unexpected screen / popup	Capture non-sensitive diagnostics and stop safely.	No blind retry.
Report returns no data	Record as no-data result, not technical success by default.	No retry until finance confirms expected behavior.
Export / file error	Mark run failed; preserve diagnostic details.	Retry export only if safe and no duplicate side effect can occur.
Notification failure	Keep report/run status intact and log notification failure.	Optional separate notification retry.

9. Logging, security and audit

- Log run ID, bot version, start/end timestamps, execution stage, outcome, output filename and sanitized error code/message.
- Never log passwords, tokens, full bank details or unnecessary vendor-sensitive information.
- Protect logs and reports using role-based access and approved retention policies.
- Use controlled development, test and production environments; promote versions through documented approval.
- Keep a clear audit trail of bot changes, test evidence, release approval and rollback version.

10. Test strategy

Test case	Expected result
Valid date range with matching records	Report runs, exports, passes checks and logs success.
Start date later than end date	Validation fails before SAP report execution.
No matching records	No-data state is logged and handled according to finance rule.
SAP login failure	Bot stops safely; no credentials appear in logs.
Unexpected popup or screen	Bot stops and records sanitized diagnostic details.
Export path unavailable	Run is marked failed; no false success notification.

Existing output filename	Approved collision policy prevents unintended overwrite.
Report period mismatch	Output is flagged and not distributed as final.
Notification service unavailable	Run status remains accurate and notification failure is logged.

11. Deployment and support

- Build and test in a non-production environment using masked or approved test data.
- Obtain Finance, SAP and RPA support sign-off before production release.
- Confirm runner/Control Room configuration, schedules, permissions, output path and support contacts.
- Document a rollback procedure: disable schedule, revert to the prior bot version and resume the approved manual process.
- Monitor early production runs and compare outputs with a manually validated report.

12. Items to confirm before implementation

- SAP transaction code, exact screen sequence and required fields.
- Date-filter semantics, timezone/business calendar and any fiscal-period rules.
- Required report columns, file format, file naming and destination.
- Whether a report with zero rows is expected, acceptable or an exception.
- Notification channel, recipients, run frequency and support escalation path.
- Approved credential vault, logging platform, retention and production access model.